Functional Autoencoders with Temporal Latent Dynamics for Forecasting

Mobina Pourmoshir, Dr. Mehdi Maadooliat
Marquette University
Department of Mathematical and Statistical Sciences

I. Background

- **Functional data are everywhere.**
  Many modern datasets consist of curves or trajectories (e.g., fertility rates, biomedical signals, energy usage).
- **Classical approach: FPCA**
  - Linear dimension reduction
  - Interpretable but limited in capturing nonlinear structure
- **Key limitation**
  - Cannot model **nonlinear dynamics, phase variation, or local deformation**
- **Motivation**
  - Learn **nonlinear, smooth, low-dimensional representations**
  - Preserve interpretability + enable **forecasting**

II. Methodology: Functional Autoencoder (FAE)

We propose a **two-stage framework**:

1) *Step 1: Nonlinear Representation Learning:* For discretized curves
[_t = (X_t(s_1), , X_t(s_J))]

- Encoder: [*t* = *E*(_t)]
- Decoder (basis representation): [*t(s)* = {m=1}^M _{tm} _m(s)]
- Reconstruction: [_t = _t]

2) *Penalized Objective:* [= _{t=1}^T |_t - _t|^2 + (X''(s))^2 ds]

- Controls:
  - **Smoothness** (denoising)
  - **Stability**

3) *Step 2: Temporal Latent Dynamics:* Latent series: [{*t*}{t=1}^T]
Modeled via VAR: [*t* = {i=1}^p A_i _{t-i} + _t]
Forecast: [*{T+1}*, , {T+L}]

4) *Step 3: Decode Forecasts:* [*{T+}(s)* = *D*(_{T+})]

III. Proposed Framework (FAE-New)

#align(center, image("images/FAE-New.pdf", width: 10in))